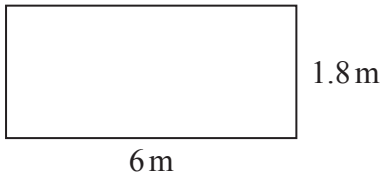
5 The diagram shows a rectangular wall.



Frank is going to cover the wall with rectangular tiles.
Each tile is 60 cm by 30 cm.

$\frac{3}{5}$ of the tiles will be white.
Some of the tiles will be green.
The rest of the tiles will be blue.

The ratio of the number of green tiles to the number of blue tiles will be 1 : 3

(a) Assuming there are no gaps between the tiles, how many tiles of each colour will Frank need?

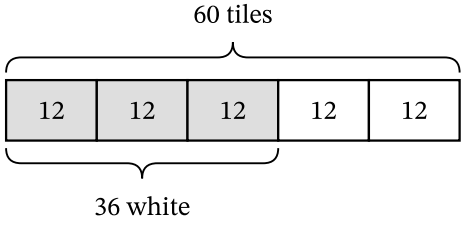
6 m = 600 cm; 1.8 m = 180 cm.

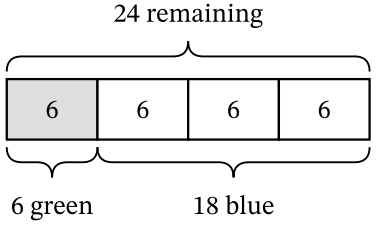
1 mark
Consistent units

Across: 600 ÷ 60 = 10; down: 180 ÷ 30 = 6.

Total = 10 × 6 = 60 tiles.

1 mark
60 tiles in total





60 ÷ 5 = 12; 3 × 12 = 36 white.

1 mark
3/5 of 60

60 − 36 = 24; 24 ÷ 4 = 6.

1 mark
Split remaining 24 into 1 + 3 parts

white tiles 36

green tiles 6

blue tiles 18

1 mark
White 36; green 6; blue 18

(5)

Frank is told that he should leave gaps between the tiles.

(b) If Frank leaves gaps between the tiles, how could this affect the number of tiles he needs?

Fewer tiles may be needed: gaps cover some of the wall.

1 mark
May need fewer

(1)

(Total for Question 5 is 6 marks)